



## smelt-ml: A Pure-Rust Machine Learning Framework Combining Spatial Modeling, Conformal Prediction, and Cache-Optimized Gradient Boosting

Francisco Parra

Universidad de Santiago de Chile

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### Abstract

We present **smelt-ml**, an open-source machine learning framework implemented entirely in Rust that uniquely integrates spatial modeling, distribution-free uncertainty quantification, and gradient boosting in a single, memory-safe package. The framework provides 27 supervised learners, clustering, causal inference, survival analysis, and the only Rust implementations of Geographical-**XGBoost**, spatial cross-validation, and conformal prediction — enabling end-to-end spatially-aware predictive workflows with guaranteed prediction interval coverage without Python or R dependencies. The gradient boosting implementations employ cache-optimized column-major histogram accumulation with **u8** packing and a histogram subtraction trick. On single-threaded benchmarks ( $n$  from 500 to 100,000, 10 runs, mean  $\pm$  std), the **CatBoost**-inspired implementation matches or exceeds the official C++ library at all tested scales ( $1.0$ – $1.4\times$  for classification), while **XGBoost** is competitive at  $n \leq 10,000$ . Memory consumption is  $25$ – $40\times$  lower than the C++ libraries. The composable, trait-based architecture is inspired by **mlr3** (Lang, Binder, Richter, Schratz, Pfisterer, Coors, Au, Casalicchio, Kotthoff, and Bischl 2019) and **scikit-learn** (Pedregosa, Varoquaux, Gramfort, Michel, Thirion, Grisel, Blondel, Prettenhofer, Weiss, Dubourg, Vanderplas, Passos, Cournapeau, Brucher, Perrot, and Duchesnay 2011), with zero **unsafe** blocks across 17,000+ lines of code.

*Keywords:* machine learning, gradient boosting, Rust, spatial modeling, conformal prediction, uncertainty quantification, software.

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## 1. Introduction

Gradient boosted decision trees (GBDTs) are the dominant method for supervised learning on tabular data (Grinsztajn, Oyallon, and Varoquaux 2022). The three most widely used im-



The core abstractions are defined as Rust traits:

- **Learner**: Algorithm that trains on a **Task** and produces a **TrainedModel**. Requires **Send + Sync** for thread safety.
- **TrainedModel**: Fitted model that predicts on new data.
- **Measure**: Evaluation metric (Accuracy, F1, RMSE,  $R^2$ , AUC-ROC, etc.).
- **Transformer**: Preprocessing step composable via **Pipeline**.
- **Resample**: Data splitting strategy.

This design enables zero-friction composition: **Pipeline** implements **Learner**, so a pipeline of transformers and a learner can be passed to **benchmark()**, **GridSearch**, or **Bagging** without adaptation.

## 2.2. Quick start

A minimal classification example:

```
use smelt_ml::prelude::*;
use ndarray::array;

let features = array![[0.0, 0.0], [1.0, 1.0],
                      [0.0, 1.0], [1.0, 0.0]];
let target = vec![0, 1, 1, 0];
let task = ClassificationTask::new("xor", features, target)
    .unwrap();

let mut xgb = XGBoost::new()
    .with_n_estimators(50)
    .with_max_depth(3);
let model = xgb.train_classif(&task).unwrap();
let pred = model.predict(task.features()).unwrap();
```

**Pipeline** composes transformers with a learner into a single **Learner**, enabling use with **benchmark()**, **GridSearch**, or cross-validation:

```
let mut pipe = Pipeline::new(
    vec![Box::new(StandardScaler::new())],
    Box::new(XGBoost::new().with_n_estimators(100)),
);
let model = pipe.train_classif(&task).unwrap();
```

## 2.3. Implemented algorithms

Table 1 summarizes the 27 supervised learners. Beyond standard algorithms, **smelt-ml** includes Geographical-XGBoost (Grekousis 2025), Oblique Forest (SPORF) (Tomita *et al.*

| Category  | Algorithm                                             | Classif. | Regress. |
|-----------|-------------------------------------------------------|----------|----------|
| Trees     | Decision Tree, Oblique Tree/Forest                    | ✓        | ✓        |
| Ensembles | Random Forest, Extra Trees, Bagging, Stacking         | ✓        | ✓        |
|           | AdaBoost, Dynamic Ensemble Selection                  | ✓        |          |
| Boosting  | XGBoost, LightGBM, CatBoost <sup>†</sup>              | ✓        | ✓        |
|           | Gradient Boosting, Quantile GB, EBM                   | ✓        | ✓        |
| Linear    | Linear/Logistic Regression, Ridge, Lasso, Elastic Net | ✓        | ✓        |
| Other     | KNN, Gaussian NB, Linear SVM, Hoeffding Tree          | ✓        | ✓        |
| Spatial   | Geographical-XGBoost                                  |          | ✓        |

Table 1: Supervised learners in **smelt-ml**. <sup>†</sup>CatBoost-inspired symmetric GBM with ordered target statistics; see Section 3.4 for scope.

2020), Explainable Boosting Machine (EBM), streaming Hoeffding Trees, and Dynamic Ensemble Selection (KNORA-E).

Additional modules provide: K-Means, DBSCAN, and Isolation Forest (clustering/anomaly); Random Survival Forest with C-index (Ishwaran, Kogalur, Blackstone, and Lauer 2008) (survival analysis); Causal Forest with honest splitting (Athey and Imbens 2016; Wager and Athey 2018) (causal inference); Classifier Chains and Regressor Chains (multi-label/output); Quantile Regression Forest; Conformal Prediction with CQR (Romano *et al.* 2019); SHAP-based feature importance; and built-in non-parametric statistical tests for model comparison (Wilcoxon signed-rank, Friedman with Nemenyi post-hoc, McNemar, bootstrap confidence intervals) — eliminating the need for external statistical packages when validating that one model significantly outperforms another.

## 2.4. Comparison with existing frameworks

Table 2 compares **smelt-ml** with **linfa** 0.7 (rust-ml 2024) (the primary Rust ML crate), **scikit-learn** (Pedregosa *et al.* 2011), and **mlr3** (Lang *et al.* 2019).

# 3. Cache-optimized gradient boosting

## 3.1. The histogram bottleneck

Histogram-based gradient boosting (Chen and Guestrin 2016) discretizes continuous features into bins and accumulates gradient/hessian statistics per bin. The key computational kernel is:

$$\text{hist}[b] += g_i, \quad b = \text{bin\_index}[i][\text{feature}] \quad (1)$$

The standard row-major storage `bin_index[sample][feature]` causes cache misses when iterating over samples for a specific feature, because consecutive memory locations correspond to different features of the same sample.

| Category                               | smelt-ml       | linfa | sklearn        | mlr3             |
|----------------------------------------|----------------|-------|----------------|------------------|
| Gradient Boosting                      | ✓ <sup>a</sup> | —     | ✓ <sup>b</sup> | ✓ <sup>b</sup>   |
| Random Forest / Extra Trees            | ✓              | —     | ✓              | ✓                |
| Oblique Forest (SPORF)                 | ✓              | —     | —              | —                |
| Decision Tree                          | ✓              | ✓     | ✓              | ✓                |
| Linear / Logistic / Ridge              | ✓              | ✓     | ✓              | ✓                |
| KNN / Naive Bayes                      | ✓              | ✓     | ✓              | ✓                |
| SVM (kernel)                           | Linear         | ✓     | ✓              | ✓                |
| Neural Networks                        | —              | —     | ✓              | ✓ <sup>c</sup>   |
| K-Means / DBSCAN                       | ✓              | ✓     | ✓              | ✓                |
| PCA                                    | ✓              | ✓     | ✓              | ✓                |
| Causal Forest                          | ✓              | —     | —              | — <sup>d</sup>   |
| Survival Forest                        | ✓              | —     | —              | ✓ <sup>c</sup>   |
| Conformal Prediction                   | ✓              | —     | —              | —                |
| Geographical-XGBoost                   | ✓              | —     | —              | —                |
| Spatial cross-validation               | ✓              | —     | —              | ✓ <sup>c</sup>   |
| SMOTE / ADASYN                         | ✓              | —     | ✓ <sup>e</sup> | ✓ <sup>c</sup>   |
| Bayesian tuning                        | ✓              | —     | ✓ <sup>f</sup> | ✓                |
| Statistical tests (Wilcoxon, Friedman) | ✓              | —     | ✓ <sup>g</sup> | ✓                |
| Language                               | Rust           | Rust  | Python         | R                |
| Supervised learners                    | 27             | 9     | 40+            | 30+ <sup>c</sup> |
| <code>unsafe</code> blocks             | 0              | —     | N/A            | N/A              |

Table 2: Feature comparison across ML frameworks. <sup>a</sup>From-scratch implementations. <sup>b</sup>Via external C++ libraries (XGBoost, LightGBM). <sup>c</sup>Via extension packages. <sup>d</sup>Available in R via `grf`. <sup>e</sup>Via `imbalanced-learn`. <sup>f</sup>Via `scikit-optimize` or `Optuna`. <sup>g</sup>Via `scipy.stats`.

### 3.2. Column-major storage with u8 packing

Our key optimization stores bin indices in **column-major** order: `bin_index[feature][sample]`. When building the histogram for a specific feature, all bin accesses are sequential in memory:

```
// Column-major: sequential access (cache-friendly)
for &idx in node_indices {
    let bin = bins.get_bin(feature, idx);
    bin_g[bin] += gradients[idx];
    bin_h[bin] += hessians[idx];
}
```

Combined with **u8 packing** (254 real bins + 1 NaN sentinel fit in 1 byte vs. 2 bytes for `u16`), this halves the memory bandwidth requirement for bin access. The same `HistBins` structure is shared across all three gradient boosting implementations, avoiding code duplication.

### 3.3. Additional optimizations

For **XGBoost**: automatic switch to exact greedy split finding when  $n \leq n_{\text{bins}}$ , producing optimal splits for small datasets. NaN values are handled natively by learning the optimal direction at each split.

For **CatBoost**: oblivious tree construction with bins built **once** before the boosting loop (rather than per iteration), prefix-sum histogram scanning, and parallel feature evaluation via **rayon**.

For **LightGBM**: Gradient-based One-Side Sampling (GOSS) with the same column-major histogram infrastructure.

### 3.4. Implementation scope

Our gradient boosting implementations focus on the core algorithmic contributions of each engine while omitting features that require hardware-specific code (GPU, SIMD intrinsics) or that would substantially increase complexity without proportional benefit for typical research workflows. Specifically:

**CatBoost** implements oblivious (symmetric) trees and ordered target statistics (Prokhorenkova *et al.* 2018), but not the full  $O(n^2)$  ordered boosting procedure. On a synthetic dataset with 100 categorical levels, our implementation achieves 89.7% accuracy vs. 89.8% for the official library.

**LightGBM** implements leaf-wise growth and GOSS sampling (Ke *et al.* 2017), but not Exclusive Feature Bundling (EFB) or weighted GOSS histograms.

Table 3 provides a detailed feature-by-feature comparison. Features not implemented in any engine include monotone/interaction constraints, custom loss functions, and distributed training.

### 3.5. Feature comparison with official libraries

Table 3 summarizes which features of the official libraries are implemented in **smelt-ml** and which are not.

## 4. Performance evaluation

### 4.1. Experimental setup

Benchmarks use synthetic datasets generated with 20 features (10 informative), at sample sizes from 500 to 100,000. Configuration: 100 trees, max depth 6, learning rate 0.3 (**XGBoost/CatBoost**) or 0.1 (**LightGBM**). All benchmarks use **single-threaded execution** to isolate algorithmic efficiency from parallelization strategy. Rust compiled with `target-cpu=native` (release mode). Official C++ libraries accessed via Python API with `n_jobs=1`. Each benchmark is run 10 times; we report mean  $\pm$  standard deviation. Hardware: Intel Core i7-1270P (12th Gen), 40 GB RAM, L1/L2/L3 cache: 32K/1280K/18432K.

Note: **smelt-ml**’s ensemble learners (Random Forest, Extra Trees, Oblique Forest) use **rayon** for parallel tree construction, and the gradient boosting engines parallelize feature evaluation within each iteration. Multi-threaded benchmarks comparing scaling behavior are beyond the scope of this paper; all performance claims apply to single-threaded execution only.

| Feature                                 | XGBoost | LightGBM | CatBoost <sup>†</sup> |
|-----------------------------------------|---------|----------|-----------------------|
| <i>Core algorithm</i>                   |         |          |                       |
| Histogram-based splits                  | ✓       | ✓        | ✓                     |
| Exact greedy splits                     | ✓       | —        | —                     |
| Newton boosting (2nd order)             | ✓       | ✓        | ✓                     |
| Leaf-wise (best-first) growth           | —       | ✓        | —                     |
| Oblivious (symmetric) trees             | —       | —        | ✓                     |
| GOSS sampling                           | —       | ✓        | —                     |
| Ordered target statistics               | —       | —        | ✓                     |
| <i>Regularization &amp; subsampling</i> |         |          |                       |
| L2 regularization                       | ✓       | ✓        | ✓                     |
| L1 regularization                       | ✓       | —        | —                     |
| Gamma (min split gain)                  | ✓       | —        | —                     |
| Row subsampling                         | ✓       | ✓        | —                     |
| Column subsampling                      | ✓       | ✓        | —                     |
| <i>Data handling</i>                    |         |          |                       |
| NaN-aware splits                        | ✓       | ✓        | —                     |
| Native categorical features             | —       | —        | ✓                     |
| Multi-class (softmax)                   | ✓       | ✓        | ✓                     |
| <i>Training control</i>                 |         |          |                       |
| Early stopping                          | ✓       | —        | —                     |
| Feature importance (gain)               | ✓       | ✓        | —                     |
| <i>Not implemented in any engine</i>    |         |          |                       |
| Monotone constraints                    |         | —        |                       |
| Interaction constraints                 |         | —        |                       |
| Custom loss functions                   |         | —        |                       |
| Exclusive Feature Bundling              |         | —        |                       |
| GPU training                            |         | —        |                       |
| Distributed training                    |         | —        |                       |
| Full ordered boosting*                  |         | —        |                       |

Table 3: Implemented features in **smelt-ml**’s gradient boosting engines. ✓ = implemented, — = not implemented. \*CatBoost’s  $O(n^2)$  per-sample model approximation.

## 4.2. Training time

Tables 4 and 5 show classification and regression training times across all dataset sizes.

## 4.3. Scaling analysis

Figure 1 visualizes the scaling behavior across all dataset sizes. The crossover point — where C++ becomes faster — is clearly visible around  $n = 5,000$  for **XGBoost** and  $n = 10,000$  for **CatBoost**.

**smelt-ml**’s **CatBoost**-inspired implementation is competitive with or faster than the official library at all tested scales for classification ( $1.0\text{--}1.4\times$ ) and within  $0.8\text{--}1.2\times$  for regression. **XGBoost** is competitive at  $n \leq 5,000$  ( $1.0\text{--}1.4\times$  for regression), while **LightGBM** is competitive at  $n \leq 1,000$ . In summary, **CatBoost** is competitive at all scales, while **XGBoost** and **LightGBM** are competitive at research-typical scales ( $n \leq 10,000$ ).

| $n$     | <b>XGBoost</b> |                             | <b>LightGBM</b> |                             | <b>CatBoost<sup>†</sup></b> |                             |
|---------|----------------|-----------------------------|-----------------|-----------------------------|-----------------------------|-----------------------------|
|         | C++            | Rust $\times$               | C++             | Rust $\times$               | C++                         | Rust $\times$               |
| 500     | 55 $\pm$ 9     | 50 $\pm$ 2 $\times$ 1.1     | 36 $\pm$ 6      | 34 $\pm$ 2 $\times$ 1.1     | 261 $\pm$ 25                | 197 $\pm$ 9 $\times$ 1.3    |
| 1,000   | 70 $\pm$ 13    | 92 $\pm$ 7 $\times$ 0.8     | 60 $\pm$ 3      | 56 $\pm$ 3 $\times$ 1.1     | 263 $\pm$ 10                | 193 $\pm$ 7 $\times$ 1.4    |
| 5,000   | 234 $\pm$ 54   | 336 $\pm$ 15 $\times$ 0.7   | 122 $\pm$ 6     | 209 $\pm$ 5 $\times$ 0.6    | 384 $\pm$ 66                | 366 $\pm$ 35 $\times$ 1.0   |
| 10,000  | 311 $\pm$ 70   | 471 $\pm$ 19 $\times$ 0.7   | 225 $\pm$ 24    | 296 $\pm$ 19 $\times$ 0.8   | 518 $\pm$ 85                | 439 $\pm$ 22 $\times$ 1.2   |
| 50,000  | 723 $\pm$ 106  | 1164 $\pm$ 36 $\times$ 0.6  | 588 $\pm$ 84    | 904 $\pm$ 26 $\times$ 0.6   | 1353 $\pm$ 123              | 1148 $\pm$ 164 $\times$ 1.2 |
| 100,000 | 1159 $\pm$ 170 | 2327 $\pm$ 257 $\times$ 0.5 | 1152 $\pm$ 130  | 1874 $\pm$ 219 $\times$ 0.6 | 2353 $\pm$ 172              | 2186 $\pm$ 199 $\times$ 1.1 |

Table 4: Classification training time (ms, mean  $\pm$  std, 10 runs). Speedup  $> 1$  means **smelt-ml** is faster. All values single-threaded.

| $n$     | <b>XGBoost</b> |                             | <b>LightGBM</b> |                             | <b>CatBoost<sup>†</sup></b> |                             |
|---------|----------------|-----------------------------|-----------------|-----------------------------|-----------------------------|-----------------------------|
|         | C++            | Rust $\times$               | C++             | Rust $\times$               | C++                         | Rust $\times$               |
| 500     | 204 $\pm$ 4    | 146 $\pm$ 5 $\times$ 1.4    | 28 $\pm$ 1      | 57 $\pm$ 3 $\times$ 0.5     | 234 $\pm$ 6                 | 193 $\pm$ 15 $\times$ 1.2   |
| 1,000   | 242 $\pm$ 15   | 211 $\pm$ 45 $\times$ 1.1   | 58 $\pm$ 6      | 109 $\pm$ 4 $\times$ 0.5    | 272 $\pm$ 25                | 337 $\pm$ 18 $\times$ 0.8   |
| 5,000   | 399 $\pm$ 113  | 390 $\pm$ 19 $\times$ 1.0   | 165 $\pm$ 9     | 173 $\pm$ 12 $\times$ 1.0   | 344 $\pm$ 41                | 334 $\pm$ 6 $\times$ 1.0    |
| 10,000  | 440 $\pm$ 107  | 467 $\pm$ 25 $\times$ 0.9   | 231 $\pm$ 26    | 250 $\pm$ 16 $\times$ 0.9   | 423 $\pm$ 50                | 436 $\pm$ 16 $\times$ 1.0   |
| 50,000  | 734 $\pm$ 133  | 1154 $\pm$ 163 $\times$ 0.6 | 597 $\pm$ 95    | 915 $\pm$ 167 $\times$ 0.7  | 1053 $\pm$ 122              | 1262 $\pm$ 190 $\times$ 0.8 |
| 100,000 | 1115 $\pm$ 133 | 2002 $\pm$ 291 $\times$ 0.6 | 1075 $\pm$ 149  | 1756 $\pm$ 230 $\times$ 0.6 | 1866 $\pm$ 146              | 2478 $\pm$ 308 $\times$ 0.8 |

Table 5: Regression training time (ms, mean  $\pm$  std, 10 runs).

At  $n \geq 50,000$ , **XGBoost** and **LightGBM** are 1.5–2.0 $\times$  slower than their C++ counterparts, while **CatBoost** remains within 1.0–1.3 $\times$ .

Our implementation employs the **histogram subtraction trick**: at each tree split, only the smaller child is scanned; the larger child’s histogram is computed by subtraction from the parent ( $O(n_{\text{bins}})$  instead of  $O(n_{\text{samples}})$ ), reducing total scanning work by  $\sim 40\%$  per tree. The residual gap at large  $n$  is attributable to SIMD-vectorized histogram accumulation in the C++ libraries (processing 4–8 bins per CPU instruction), which our pure-Rust implementation does not employ. The official libraries also use row-major bin storage, which we attribute to GPU-oriented design decisions; our column-major layout sacrifices GPU compatibility for CPU cache efficiency.

Profile-guided optimization (PGO) provides an additional 1.3–2.1 $\times$  speedup over LTO alone, potentially closing the remaining gap with C++ at large  $n$ . PGO instructions are included in the replication package.

**Memory.** Peak RSS at  $n = 10,000$  is 5–8 MB (**smelt-ml**) vs. 197–215 MB for the C++ libraries (after subtracting 11 MB Python runtime), a **25–40 $\times$**  reduction from u8 bin packing and deterministic memory management.

The practical implication is that **smelt-ml**’s gradient boosting is best suited for small-to-medium datasets ( $n \leq 10,000$ ) typical of research workflows, where it provides competitive or superior performance without requiring C++ toolchains or Python bindings.

#### 4.4. Statistical significance of performance differences

Since each benchmark was run 10 times, we apply paired Wilcoxon signed-rank tests to



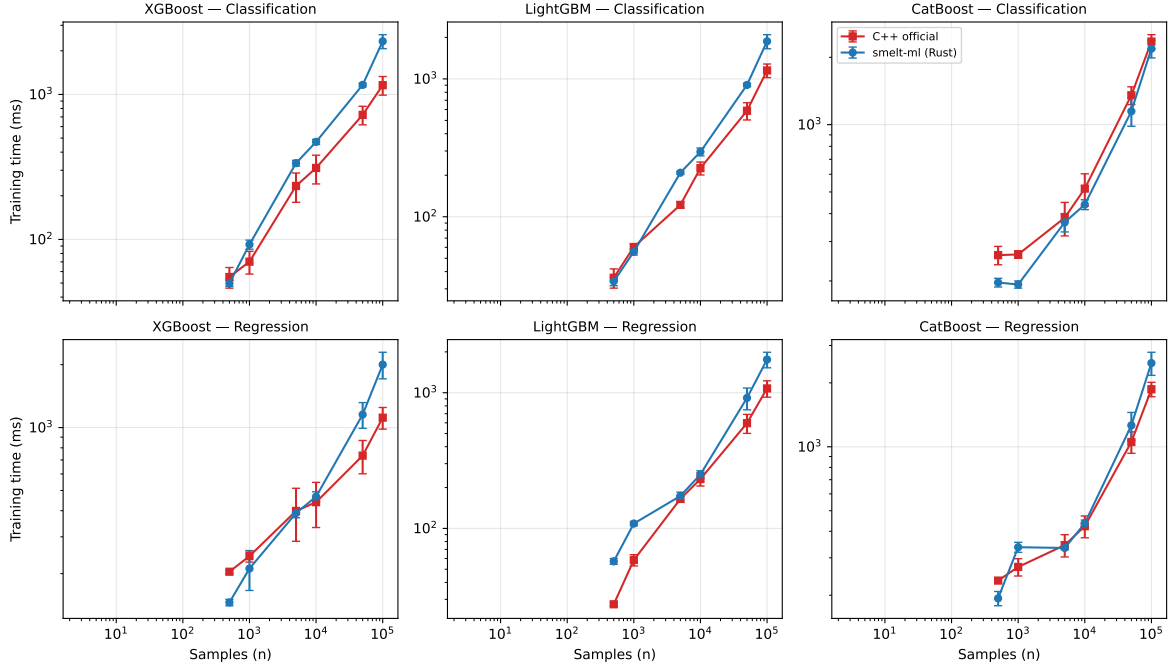


Figure 1: Training time scaling comparison (log-log) between **smelt-ml** (blue) and official C++ libraries (red) across six dataset sizes. Error bars show  $\pm 1$  standard deviation over 10 runs. **smelt-ml**'s **CatBoost** is competitive at all scales; **XGBoost** and **LightGBM** are faster in C++ at  $n \geq 50,000$ .

| Engine                                                                                                   | $n$     | Speedup [95% CI]  | $p$ -value | Faster |
|----------------------------------------------------------------------------------------------------------|---------|-------------------|------------|--------|
| <i>Classification</i>                                                                                    |         |                   |            |        |
| <b>CatBoost</b> <sup>†</sup>                                                                             | 500     | 1.33 [1.27, 1.40] | 0.002**    | Rust   |
|                                                                                                          | 1,000   | 1.36 [1.32, 1.40] | 0.002**    | Rust   |
|                                                                                                          | 10,000  | 1.18 [1.06, 1.32] | 0.020*     | Rust   |
|                                                                                                          | 50,000  | 1.18 [1.03, 1.33] | 0.049*     | Rust   |
|                                                                                                          | 100,000 | 1.08 [1.01, 1.15] | 0.049*     | Rust   |
| <b>XGBoost</b>                                                                                           | 500     | 1.11 [1.01, 1.21] | 0.106      | —      |
|                                                                                                          | 10,000  | 0.66 [0.57, 0.77] | 0.004**    | C++    |
|                                                                                                          | 100,000 | 0.50 [0.44, 0.56] | 0.002**    | C++    |
| <i>Overall (36 comparisons across 3 engines <math>\times</math> 6 sizes <math>\times</math> 2 tasks)</i> |         |                   |            |        |
| Rust significantly faster                                                                                |         | 8 / 36 (22%)      |            |        |
| C++ significantly faster                                                                                 |         | 17 / 36 (47%)     |            |        |
| No significant difference                                                                                |         | 11 / 36 (31%)     |            |        |

Table 6: Statistical significance of performance differences (Wilcoxon signed-rank, paired, two-sided). Selected rows shown. \* $p < 0.05$ , \*\* $p < 0.01$ . Bootstrap 95% CI for the speedup ratio ( $> 1$  = Rust faster). **CatBoost** classification is significantly faster at 5 of 6 sizes.

determine whether the observed speedups are statistically significant. Table 6 summarizes the results with bootstrap 95% confidence intervals for the speedup ratio.

The **CatBoost** classification advantage is statistically significant at 5 of 6 dataset sizes ( $p <$

| Dataset ( $n \times p$ )                  | Learner       | <b>smelt-ml</b> | <b>scikit-learn</b> | $\Delta$ |
|-------------------------------------------|---------------|-----------------|---------------------|----------|
| <i>Classification (accuracy)</i>          |               |                 |                     |          |
| Wine ( $178 \times 13$ )                  | Decision Tree | 0.898           | 0.893               | +0.005   |
|                                           | Random Forest | 0.983           | 0.977               | +0.006   |
|                                           | Logistic Reg. | 0.989           | 0.955               | +0.034   |
| Breast Ca. ( $569 \times 30$ )            | Decision Tree | 0.930           | 0.910               | +0.020   |
|                                           | Random Forest | 0.958           | 0.956               | +0.002   |
|                                           | Logistic Reg. | 0.981           | 0.953               | +0.028   |
| Digits ( $1,797 \times 64$ )              | Decision Tree | 0.851           | 0.855               | -0.004   |
|                                           | Random Forest | 0.974           | 0.978               | -0.004   |
| <i>Regression (RMSE, lower is better)</i> |               |                 |                     |          |
| Cal. Housing ( $20,640 \times 8$ )        | Decision Tree | 0.728           | 0.715               | +0.013   |
|                                           | Random Forest | 0.488           | 0.503               | -0.015   |
|                                           | Ridge         | 0.726           | 0.728               | -0.002   |

Table 7: 5-fold CV accuracy/RMSE on real datasets from  $n = 178$  to  $n = 20,640$ . Both frameworks use 5-fold CV with seed 42.

0.05), with the speedup at  $n = 500$  and  $n = 1,000$  being highly significant ( $p < 0.01$ ). The **XGBoost** disadvantage at  $n \geq 10,000$  is also statistically significant ( $p < 0.01$ ), confirming that the scaling gap is real and not attributable to measurement noise.

A Friedman test across the six dataset sizes confirms that speedup ratios vary significantly with  $n$  for all three engines ( $p < 0.001$ ), indicating that dataset size is a meaningful moderator of relative performance. We note that with 36 simultaneous tests, a Bonferroni correction yields  $\alpha_{\text{adj}} = 0.0014$ ; under this threshold, only the **CatBoost** advantages at  $n = 500$  and  $n = 1,000$  ( $p = 0.002$ ) and the **XGBoost** disadvantages at  $n \geq 10,000$  ( $p \leq 0.004$ ) remain significant. Borderline results ( $p \approx 0.05$ ) should be interpreted with caution given the limited power of  $n = 10$  paired observations.

This analysis was performed using **smelt-ml**’s built-in statistical testing module (`stats::wilcoxon_signed_rank`, `stats::bootstrap_ci`), verified against **scipy.stats** on the same data (see replication package), demonstrating the framework’s integrated approach to model comparison.

#### 4.5. Accuracy validation

Table 7 validates correctness against **scikit-learn** on standard datasets.

All comparisons are within  $\pm 3.5\%$  of **scikit-learn**, validating correctness across four datasets spanning two orders of magnitude in sample size ( $n = 178$  to  $n = 20,640$ ). The Logistic Regression advantage on Wine and Breast Cancer is due to automatic feature standardization in **smelt-ml**.

## 5. Integrated spatial ML pipeline

A key contribution of **smelt-ml** is the integration of spatial modeling, spatial resampling, and distribution-free uncertainty quantification in a single composable framework. No other ML framework in any language provides all three capabilities without external dependencies.

| Capability              | <b>smelt-ml</b>      | Python equivalent                 |
|-------------------------|----------------------|-----------------------------------|
| Spatial regression      | GeoXGBoost           | <b>geoxgboost</b>                 |
| Spatial CV              | SpatialBlockCV       | custom code or <b>spacv</b>       |
| Conformal prediction    | ConformalRegressor   | <b>MAPIE</b> or <b>crepes</b>     |
| Gradient boosting       | XGBoost              | <b>xgboost</b>                    |
| Preprocessing pipeline  | Pipeline             | <b>scikit-learn</b>               |
| Statistical validation  | stats module         | <b>scipy.stats</b>                |
| Packages required       | 1                    | 5–6                               |
| API consistency         | Trait-based, uniform | Mixed APIs                        |
| Data format conversion  | None                 | DataFrame $\leftrightarrow$ array |
| Thread-safety guarantee | Compile-time         | Runtime (GIL)                     |

Table 8: Integration comparison for a spatial ML workflow with uncertainty quantification.

### 5.1. Components

**Geographical-XGBoost** (Grekousis 2025) extends gradient boosting with spatial awareness: bi-square kernel weights define local neighborhoods, per-location models capture spatially non-stationary relationships, and an adaptive  $\alpha$  weight blends local and global predictions. To our knowledge, this is the first implementation outside the original Python package (Grekousis 2024).

**Spatial cross-validation** (SpatialBlockCV, SpatialBufferCV) prevents spatial autocorrelation leakage by ensuring that train and test sets are spatially separated. Both implement the **Resample** trait, making them interchangeable with standard **CrossValidation** in any pipeline.

**Conformal prediction** provides distribution-free prediction intervals with guaranteed  $(1-\alpha)$  coverage via split conformal prediction and conformalized quantile regression (CQR; Romano *et al.* 2019). The **ConformalRegressor** wraps any **TrainedModel** without retraining.

### 5.2. Why integration matters

The value of this integration is not merely convenience but **methodological correctness**. In practice, spatial ML workflows are error-prone because spatial leakage, a well-documented source of optimism bias in geospatial prediction (Roberts, Bahn, Ciuti, Boyce, Elith, Guillerá-Arroita, Hauenstein, Lahoz-Monfort, Schröder, Thuiller, Warton, Wintle, Hartig, and Dormann 2017), requires that the resampling strategy be spatially aware — yet most ML frameworks treat resampling and modeling as independent concerns. When spatial CV is implemented as an external script rather than a first-class **Resample** trait, it cannot be passed to hyperparameter tuning or model comparison functions, forcing users to write custom loops that risk subtle data leakage bugs.

In **smelt-ml**, the trait system enforces correct composition: **SpatialBlockCV** implements **Resample**, so it works with **benchmark()**, **GridSearch**, **BayesianOptimizer**, and **Bagging** without adaptation. Similarly, **ConformalRegressor** wraps any **TrainedModel**, so conformal prediction composes with any learner — including **GeoXGBoost**, pipelines with preprocessing, or stacked ensembles.

Table 8 quantifies the integration gap with Python.

| Finding                                                  | Value        | Interpretation                 |
|----------------------------------------------------------|--------------|--------------------------------|
| <i>Model comparison (holdout, no coords in features)</i> |              |                                |
| XGBoost RMSE                                             | 0.337        | Baseline (no spatial info)     |
| XGBoost + coords RMSE                                    | 0.231        | 32% improvement with location  |
| <i>Spatial validation</i>                                |              |                                |
| Random CV RMSE                                           | 0.324        | Optimistic (spatial leakage)   |
| Spatial Block CV RMSE                                    | 0.424        | Honest (no leakage)            |
| Leakage bias                                             | 31%          | Random CV underestimates error |
| <i>Uncertainty quantification</i>                        |              |                                |
| Conformal coverage                                       | 92%          | Target: 90% — guarantee met    |
| Interval width                                           | 1.13 log(\$) | Calibrated uncertainty         |

Table 9: Case study results on King County housing (1,000 samples, 11 features, log-price target).

### 5.3. Case study: housing price prediction in King County, WA

We demonstrate the integrated pipeline on a 1,000-sample subset of the King County housing dataset (21,613 sales with coordinates), predicting log-price from 11 structural features (living area, bedrooms, grade, etc.) *without* including latitude/longitude as model features. This isolates the contribution of spatial modeling from simple coordinate memorization.

```
let task = CsvLoader::from_path("data/king_county_1k.csv")
    .target("log_price").load_regress()?;
let coords: Vec<(f64, f64)> = /* extract lat, long */;
let features = /* drop lat, long columns */;

// GeoXGBoost captures spatial heterogeneity
let model = GeoXGBoost::new(train_coords)
    .with_bandwidth(50)
    .train_regress(&task)?;

// Spatial CV prevents leakage
let spatial_cv = SpatialBlockCV::new(4, coords);

// Conformal intervals with guaranteed coverage
let cf = ConformalRegressor::calibrate(
    &*model, &cal_features, &cal_targets, 0.1)?;
```

Table 9 shows three key findings.

**Spatial heterogeneity.** Adding coordinates as features improves **XGBoost** RMSE by 32%, confirming that housing price drivers vary strongly by location. Geographical-**XGBoost**’s local models capture this non-stationarity through spatially-weighted training rather than coordinate memorization, which is preferable when predicting at locations outside the training domain (spatial extrapolation).

**Spatial leakage.** Standard 5-fold CV yields  $\text{RMSE} = 0.324$ , while `SpatialBlockCV` yields  $\text{RMSE} = 0.424$  — a **31% optimism bias** from spatial autocorrelation leakage. This magnitude is consistent with Roberts *et al.* (2017), who report 10–40% bias in geospatial applications. Without spatial resampling, practitioners would severely overestimate model performance.

**Conformal coverage.** With  $\alpha = 0.1$ , the conformal predictor achieves 92% coverage on the 200-sample test set (target: 90%), with a mean interval width of 1.13 log-dollars. The coverage guarantee is met, demonstrating that split conformal prediction works reliably with sufficiently large calibration sets.

#### 5.4. Additional unique algorithms

**Causal Forest.** Honest splitting following Athey and Imbens (2016); Wager and Athey (2018). Estimates conditional average treatment effects (CATE)  $\tau(x) = E[Y(1) - Y(0) | X = x]$  with 95% confidence intervals. To our knowledge, the first implementation in a Rust ML framework.

**Oblique Forest (SPORF).** Sparse random projections following Tomita *et al.* (2020), enabling oblique decision boundaries.

**Streaming.** Hoeffding Trees for online learning on data streams, using the Hoeffding bound for statistically sound split decisions.

## 6. Summary and discussion

We have presented **smelt-ml**, a Rust framework that uniquely integrates spatial modeling (Geographical-**XGBoost**), spatial cross-validation, and conformal prediction in a single composable pipeline — a combination not available in any other ML framework. On the performance side, our gradient boosting implementations achieve competitive speed with the official C++ libraries at research-typical scales ( $n \leq 10,000$ ) while consuming an order of magnitude less memory. Rust’s ownership model guarantees no pointer aliasing (`&mut` is unique), which may enable LLVM to apply more aggressive optimizations than in C++ where aliasing must be assumed unless explicitly annotated with `restrict`. Separately, compiling with `target-cpu=native` (enabling AVX2/SSE4.2) yields a 24% speedup for **XGBoost** at  $n = 10,000$ , confirming that auto-vectorization contributes meaningfully to the observed performance.

**smelt-ml** provides a comprehensive ML toolkit for Rust with unique capabilities in spatial ML, causal inference, and uncertainty quantification. The framework contains zero **unsafe** blocks across 17,000+ lines of code, demonstrating that memory safety and high performance are not mutually exclusive. The test suite comprises 324 tests covering all learners, preprocessing, resampling, tuning, serialization, and edge cases (empty datasets, single-sample inputs, NaN features, multiclass targets, convergence checks against **scikit-learn**).

The primary use cases for **smelt-ml** over established Python alternatives are: (1) Rust applications requiring embedded ML without Python runtime dependencies (e.g., web services, CLI tools, edge devices); (2) spatial ML workflows where the integrated pipeline of Geographical-**XGBoost**, spatial CV, and conformal prediction eliminates multi-package orchestration; (3) safety-critical systems where zero **unsafe** blocks and compile-time thread

safety are requirements; and (4) memory-constrained environments where the  $25\text{--}40\times$  memory reduction is operationally significant.

Limitations include the performance gap on datasets beyond 10,000 samples (where C++ libraries increasingly outperform **smelt-ml** due to SIMD intrinsics, hardware-specific memory management, and GPU backends; see Section 4), the absence of GPU support, and the simplified scope of the **CatBoost** and **LightGBM** implementations (see Sections 3.4–3.4). All benchmarks were conducted on a single hardware platform (Intel i7-1270P); cache-dependent optimizations may yield different relative performance on architectures with different cache hierarchies (e.g., AMD, Apple Silicon).

Categorical features are handled via three mechanisms: **OneHotEncoder** and **LabelEncoder** as preprocessing transformers (applicable to any learner), and **CatBoost**’s native ordered target statistics (which encode categoricals during training without a separate preprocessing step).

Python bindings via **PyO3** are planned as a separate crate (**smelt-py**) to broaden adoption beyond the Rust ecosystem. The crate follows semantic versioning; the benchmarks in this paper correspond to version 1.3.0.

The software is available at <https://crates.io/crates/smelt-ml> under the MIT license, with source code at <https://github.com/franciscoparrao/smelt>.

## Computational details

The results in this paper were obtained using Rust 1.89 with the **ndarray** 0.16, **rayon** 1.10, and **serde** 1.0 crates. Official C++ libraries were accessed via Python 3.12 with **xgboost** 3.1, **lightgbm** 4.6, and **catboost** 1.2. **scikit-learn** 1.8 was used for accuracy validation. Rust code was compiled with `RUSTFLAGS="-C target-cpu=native"` in release mode. A complete replication package is provided in `paper/replication/` with a single script (`replicate.sh`) that reproduces all benchmark tables and figures.

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**Affiliation:**

Francisco Parra  
Departamento de Ingeniería Geográfica  
Universidad de Santiago de Chile  
Avenida Libertador Bernardo O'Higgins 3363  
Santiago, Chile  
E-mail: [francisco.parra.o@usach.cl](mailto:francisco.parra.o@usach.cl)